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$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(1 + \frac{5x+6}{2x^2+3x+4} \right)^{7x+8} \\ &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{5x+6}{2x^2+3x+4} \right)^{\frac{2x^2+3x+4}{5x+6} \cdot \frac{5x+6}{2x^2+3x+4}} \right)^{7x+8} \\ &= e^{\lim_{x \rightarrow \infty} \frac{(5x+6)(7x+8)}{2x^2+3x+4}} \\ &= e^{\lim_{x \rightarrow \infty} \frac{35x^2+82x+48}{2x^2+3x+4}} \\ &= e^{\lim_{x \rightarrow \infty} \frac{35x^2}{2x^2}} \\ &= e^{\frac{35}{2}} \end{aligned}$$

References